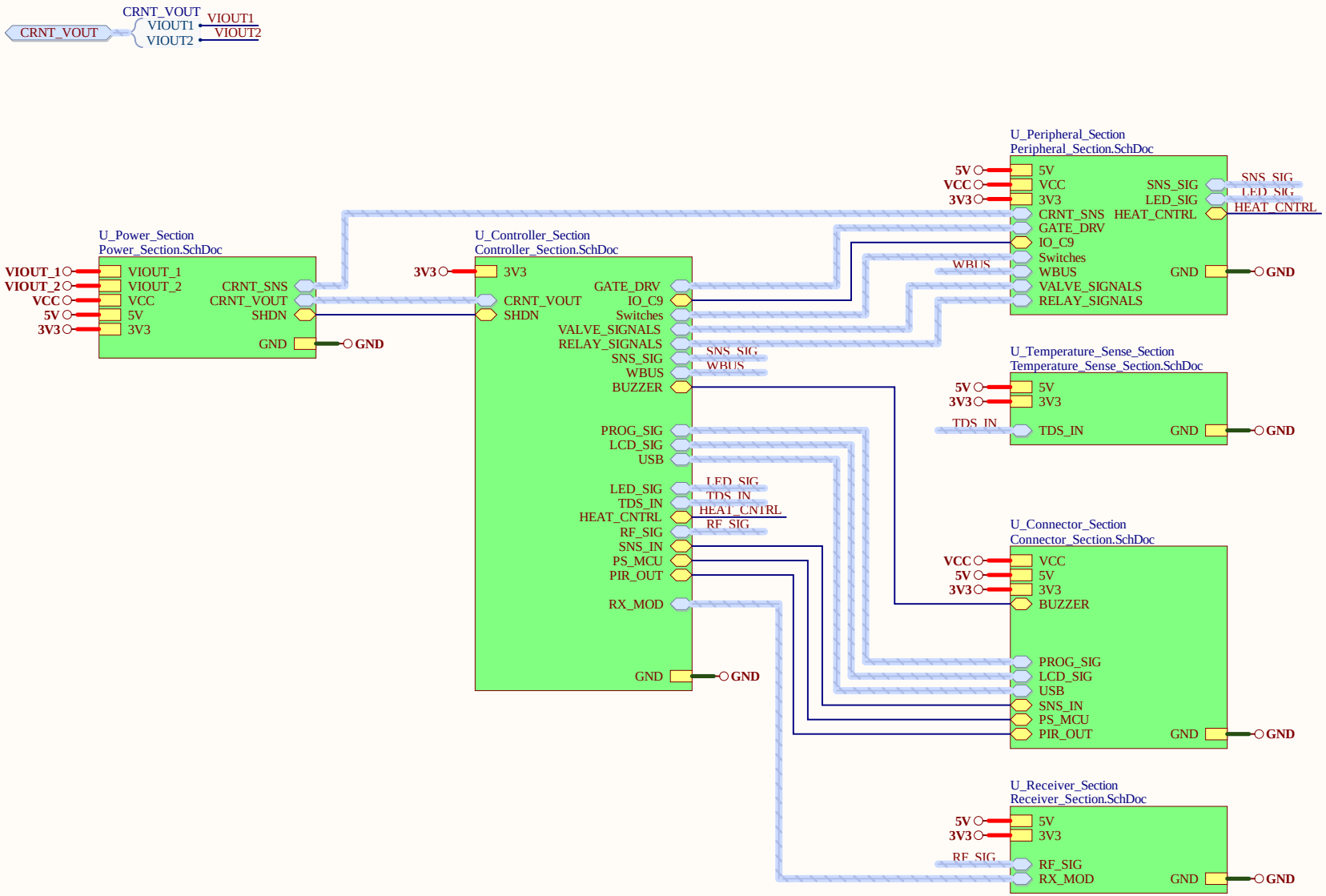
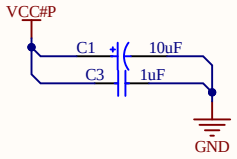

CRNT_VOUT
CRNT_VOUT
VIOUT1
VIOUT2
VIOUT1
VIOUT2
U_Power_Section
Power_Section.SchDoc
VIOUT_1
VIOUT_2
VCC
5V
3V3
VIOUT_1
VIOUT_2
VCC
5V
3V3
CRNT_SNS
CRNT_VOUT
SHDN
GND
GND
U_Controller_Section
Controller_Section.SchDoc
3V3
3V3
CRNT_VOUT
SHDN
GATE_DRV
IO_C9
Switches
VALVE_SIGNALS
RELAY_SIGNALS
SNS_SIG
WBUS
BUZZER
PROG_SIG
LCD_SIG
USB
LED_SIG
TDS_IN
HEAT_CNTRL
RF_SIG
SNS_IN
PS_MCU
PIR_OUT
RX_MOD
GND
GND
U_Peripheral_Section
Peripheral_Section.SchDoc
5V
VCC
3V3
CRNT_SNS
GATE_DRV
IO_C9
Switches
WBUS
VALVE_SIGNALS
RELAY_SIGNALS
SNS_SIG
LED_SIG
HEAT_CNTRL
GND
GND
U_Temperature_Sense_Section
Temperature_Sense_Section.SchDoc
5V
3V3
TDS_IN
5V
3V3
TDS_IN
GND
GND
U_Connector_Section
Connector_Section.SchDoc
VCC
5V
3V3
BUZZER
PROG_SIG
LCD_SIG
USB
SNS_IN
PS_MCU
PIR_OUT
VCC
5V
3V3
BUZZER
PROG_SIG
LCD_SIG
USB
SNS_IN
PS_MCU
PIR_OUT
GND
GND
U_Receiver_Section
Receiver_Section.SchDoc
5V
3V3
RF_SIG
5V
3V3
RF_SIG
RX_MOD
GND
GND

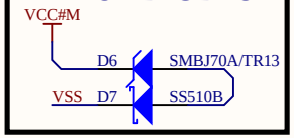

1
2
3
4
A
B
C
D
CRNT_SNS
CRNT_SNS
SHDN
SHDN
CRNT_VOUT
CRNT_VOUT
U_Regulator_Section
Regulator_Section.SchDoc
VCC#5
5V
3V3
VCC#5
5V
3V3
GND
GND
U_Current_Sense_Section
Current_Sense_Section.SchDoc
VIOUT_1
VIOUT_2
VCC#P
5V
VIOUT_1
VIOUT_2
VCC#P
5V
CRNT_VOUT
CRNT_VOUT
CRNT_SNS
CRNT_SNS
GND
GND
U_Reverse Protection_Section
Reverse Protection_Section.SchDoc
VCC#P
VCC
VCC#P
VCC
SHDN
SHDN
GND
GND
U_Voltage_Monitoring_Section
Voltage_Monitoring_Section.SchDoc
VCC#P
VCC#5
VCC#P
VCC#5
GND
GND
VCC
5V
3V3
VIOUT_1
VIOUT_2
GND
VCC
5V
3V3
VIOUT_1
VIOUT_2
GND



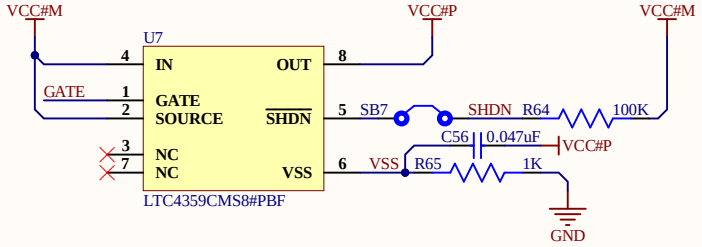
DECOUPLING CAPACITOR



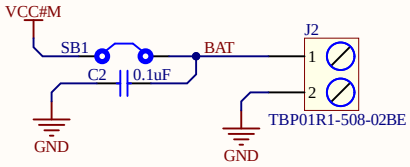
INPUT DIODES



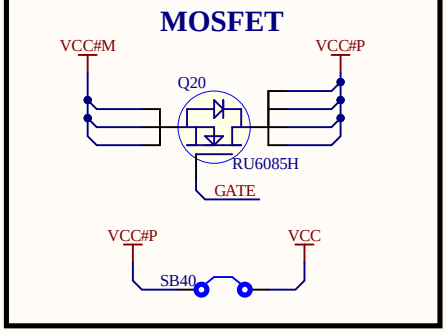
REVERSE POLARITY PROTECTION IC

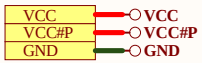


BATTERY / MAIN INPUT

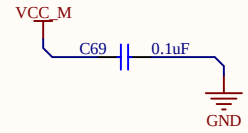


MOSFET

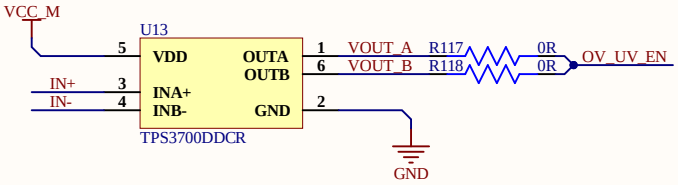




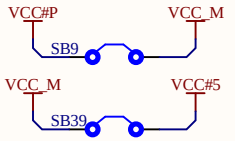
DECOUPLING CAPACITOR



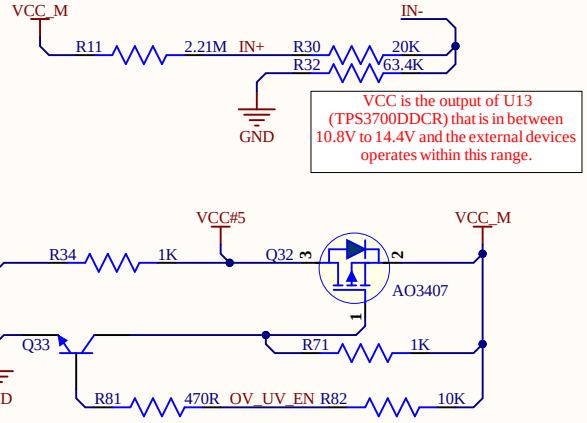
UNDER AND OVER VOLTAGE PROTECTION IC

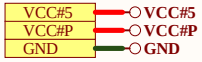


VOLTAGE TO MONITOR

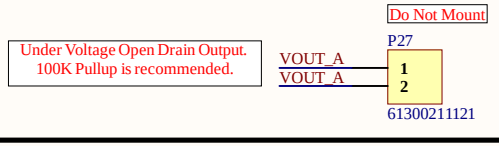


SUPPORTING CIRCUITRY

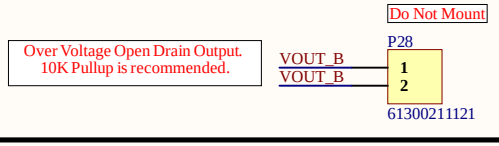


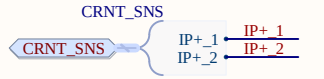


UNDER VOLTAGE OUTPUT

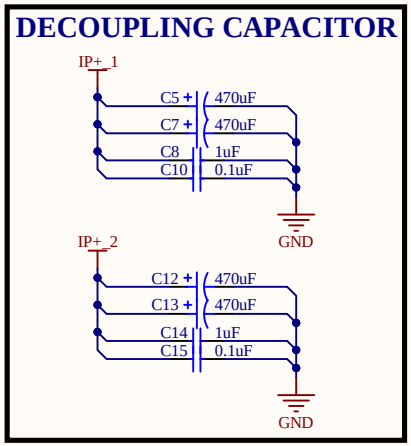


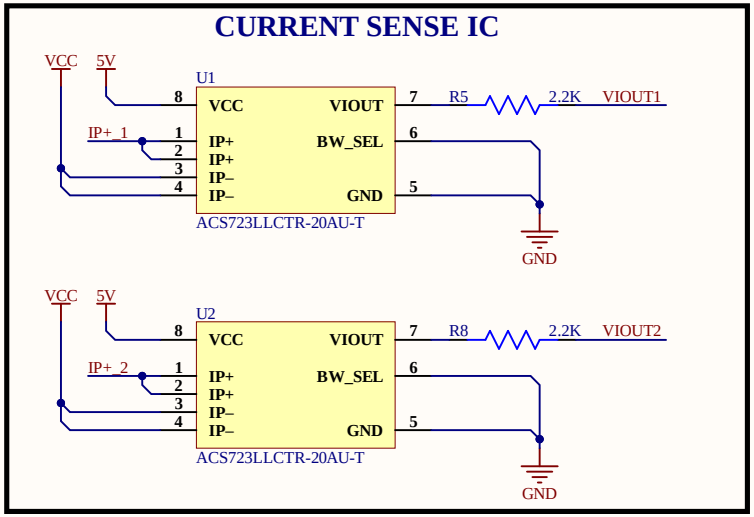
OVER VOLTAGE OUTPUT

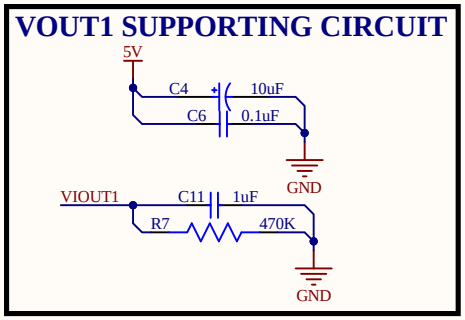


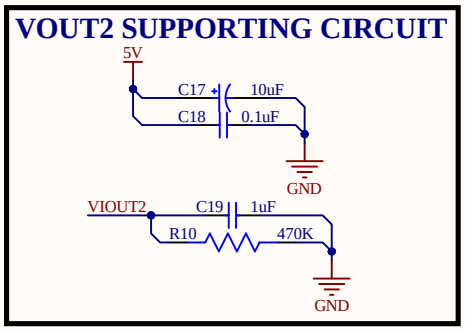




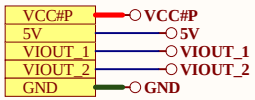




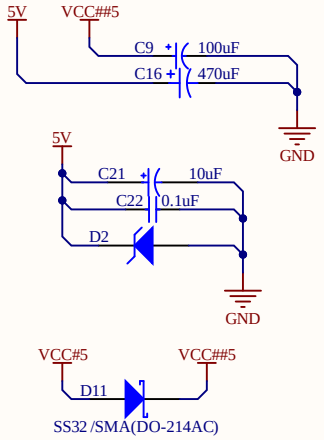




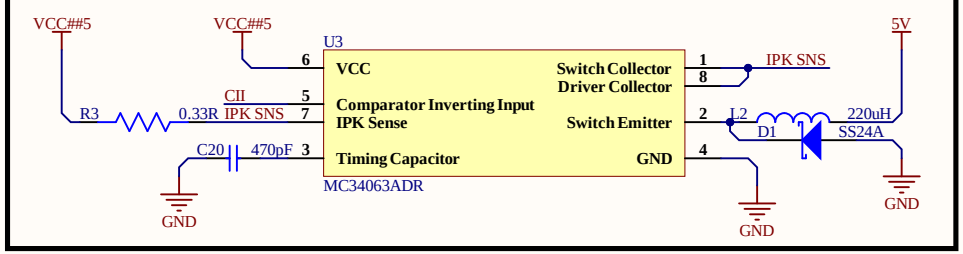
Two current sensing IC's are used to provide support to monitor current of two Pumps. Whereas currently we have only one pump in our V3 Controller.



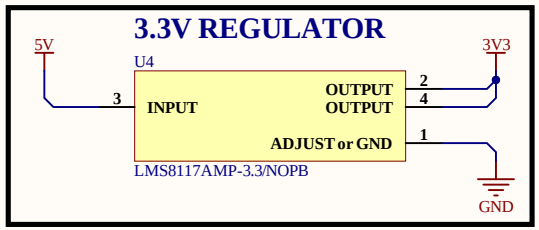
DECOUPLING CAPACITOR



5V BUCK REGULATOR

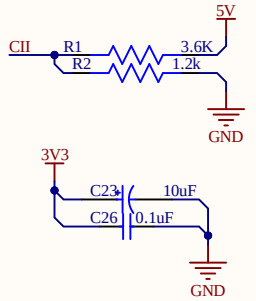


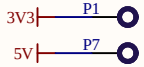
3.3V REGULATOR

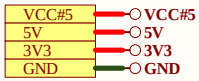


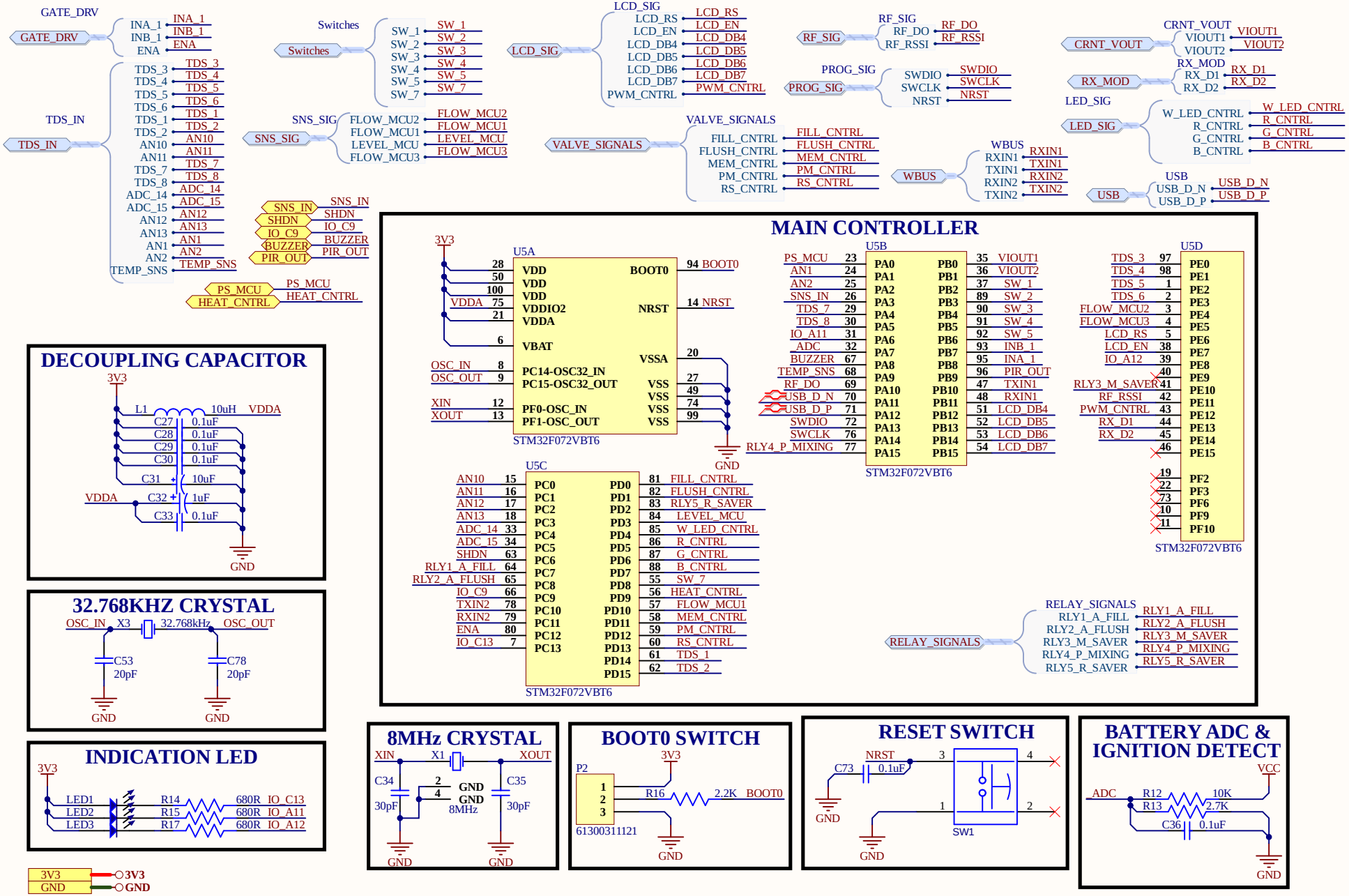
VOUT = 1.25 (1+ R2/R1)
R2=3.6K
R1=1.2K

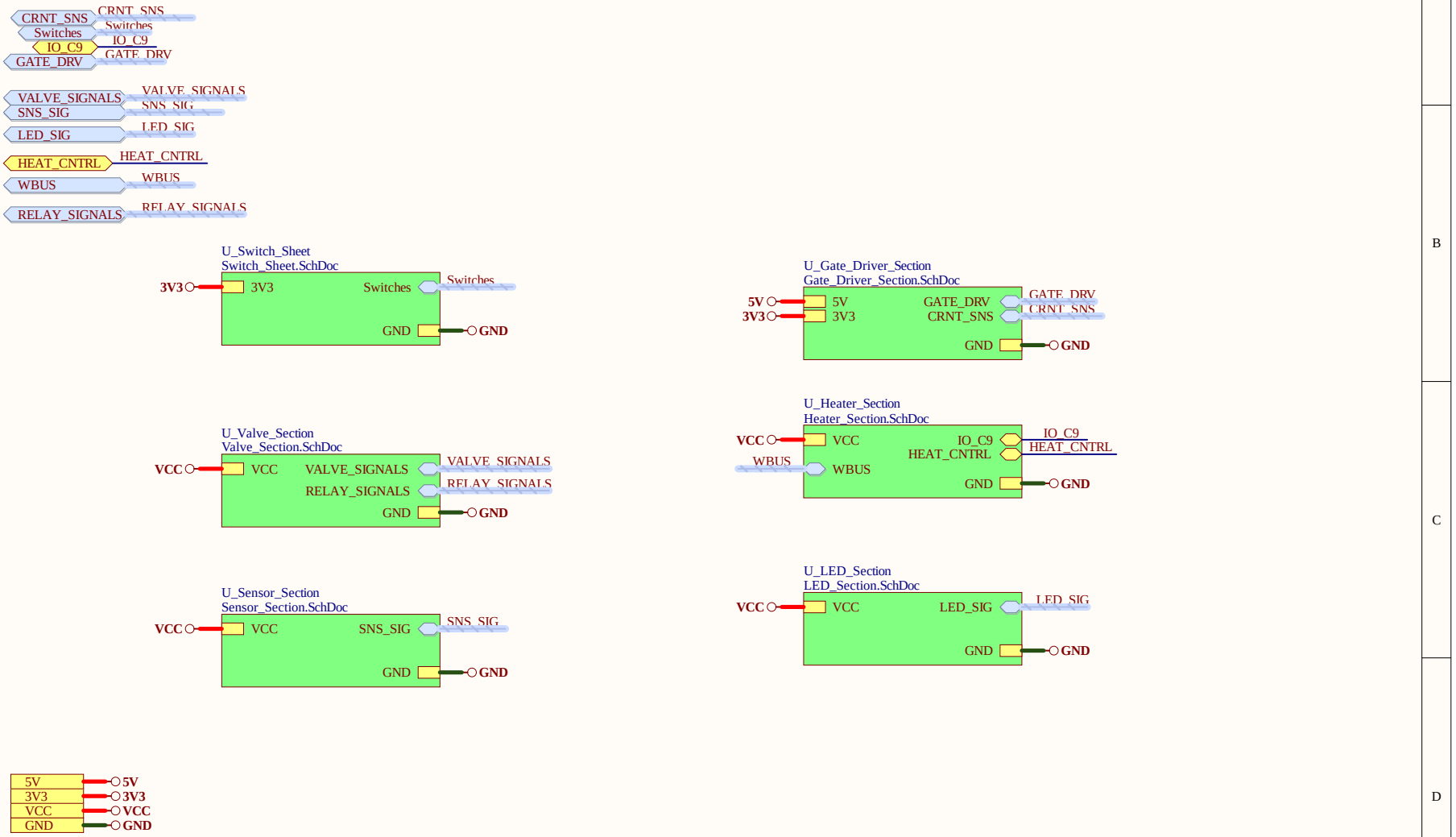
SUPPORTING CIRCUIT

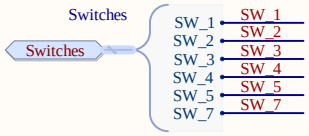


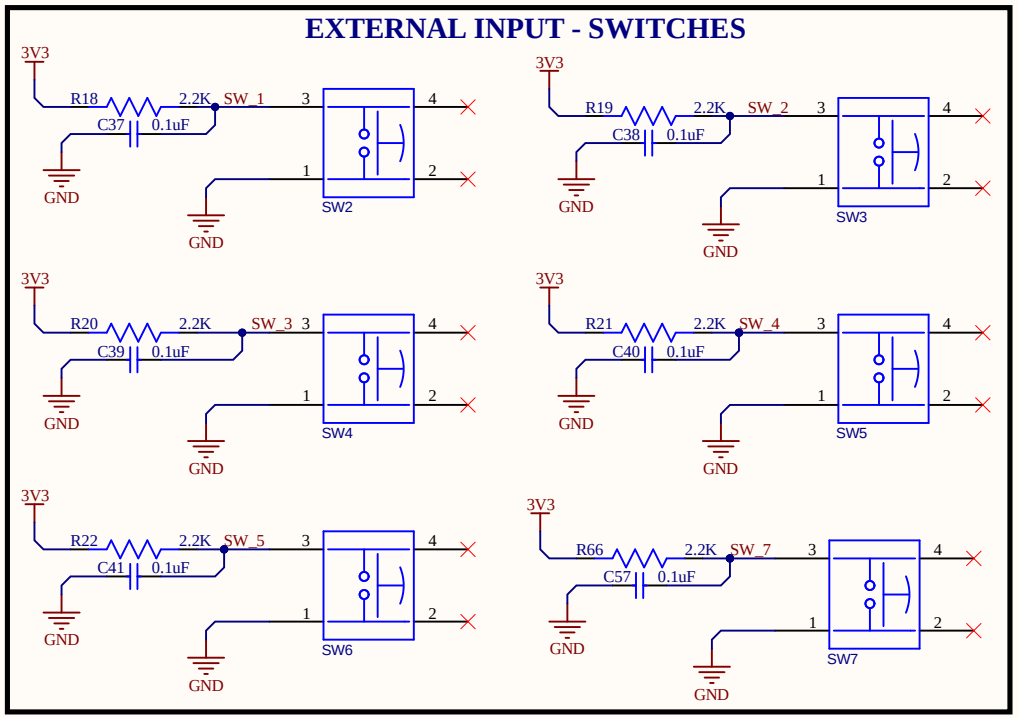


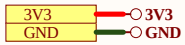


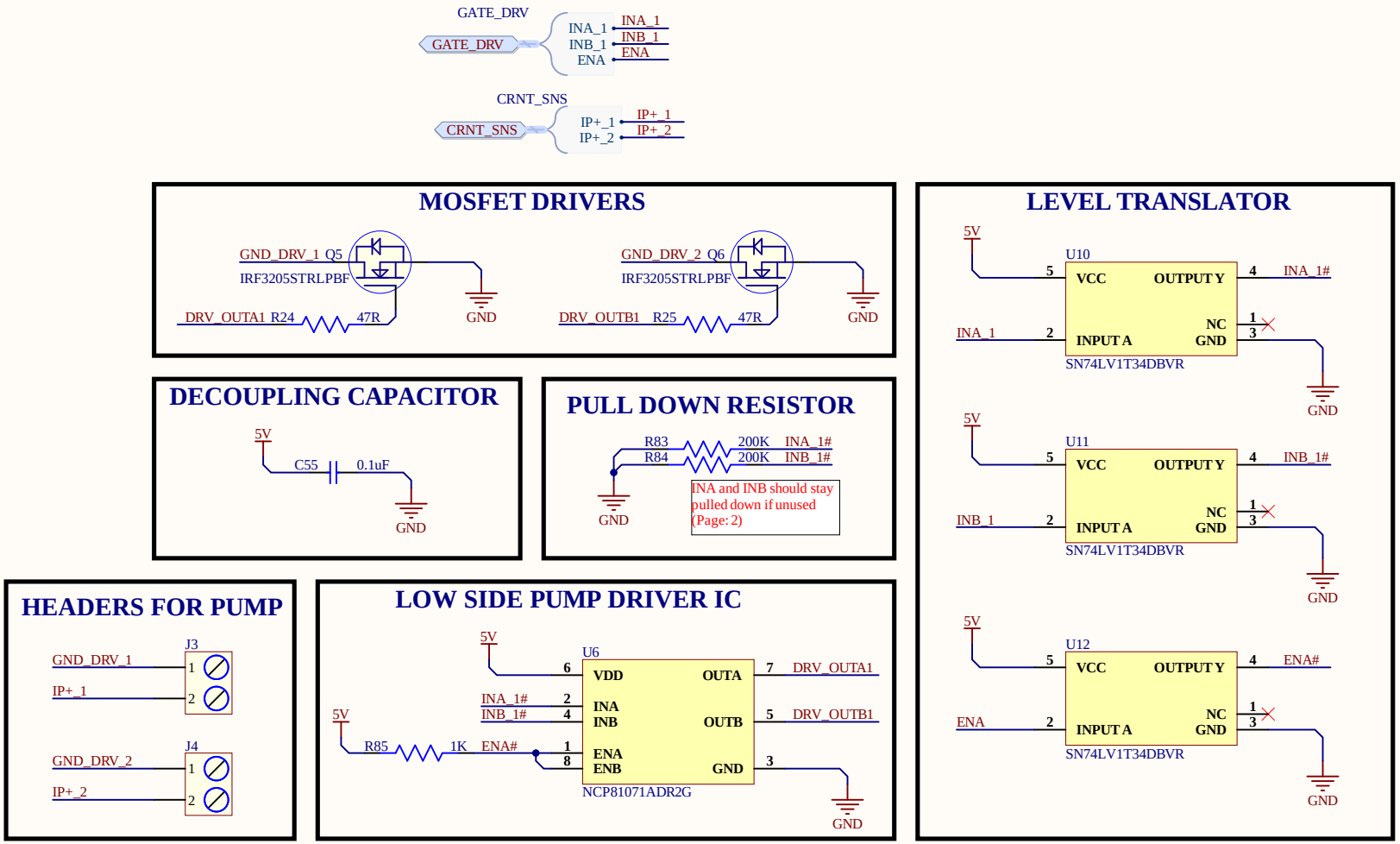

GATE_DRV
INA_1
INB_1
ENA
TDS_IN
TDS_3
TDS_4
TDS_5
TDS_6
TDS_1
TDS_2
AN10
AN11
TDS_7
TDS_8
ADC_14
ADC_15
AN12
AN13
AN1
AN2
TEMP_SNS
Switches
SW_1
SW_2
SW_3
SW_4
SW_5
SW_7
SNS_SIG
FLOW_MCU2
FLOW_MCU1
LEVEL_MCU
FLOW_MCU3
LCD_SIG
LCD_RS
LCD_EN
LCD_DB4
LCD_DB5
LCD_DB6
LCD_DB7
PWM_CNTRL
RF_SIG
RF_DO
RF_RSSI
PROG_SIG
SWDIO
SWCLK
NRST
VALVE_SIGNALS
FILL_CNTRL
FLUSH_CNTRL
MEM_CNTRL
PM_CNTRL
RS_CNTRL
WBUS
RXIN1
TXIN1
RXIN2
TXIN2
CRNT_VOUT
VIOUT1
VIOUT2
RX_MOD
RX_D1
RX_D2
LED_SIG
W_LED_CNTRL
R_CNTRL
G_CNTRL
B_CNTRL
USB
USB_D_N
USB_D_P
SNS_IN
SHDN
IO_C9
BUZZER
PIR_OUT
PS_MCU
HEAT_CNTRL
MAIN CONTROLLER
U5A
STM32F072VBT6
U5B
U5C
U5D
DECOUPLING CAPACITOR
L1 10uH VDDA
C27 0.1uF
C28 0.1uF
C29 0.1uF
C30 0.1uF
C31 10uF
C32 1uF
C33 0.1uF
32.768KHZ CRYSTAL
X3 32.768kHz
C53 20pF
C78 20pF
INDICATION LED
LED1 R14 680R IO_C13
LED2 R15 680R IO_A11
LED3 R17 680R IO_A12
8MHz CRYSTAL
XIN X1 XOUT
C34 30pF
C35 30pF
8MHz
BOOT0 SWITCH
P2
R16 2.2K BOOT0
61300311121
RESET SWITCH
NRST
C73 0.1uF
SW1
BATTERY ADC & IGNITION DETECT
ADC
R12 10K
R13 2.7K
C36 0.1uF
VCC
3V3
GND

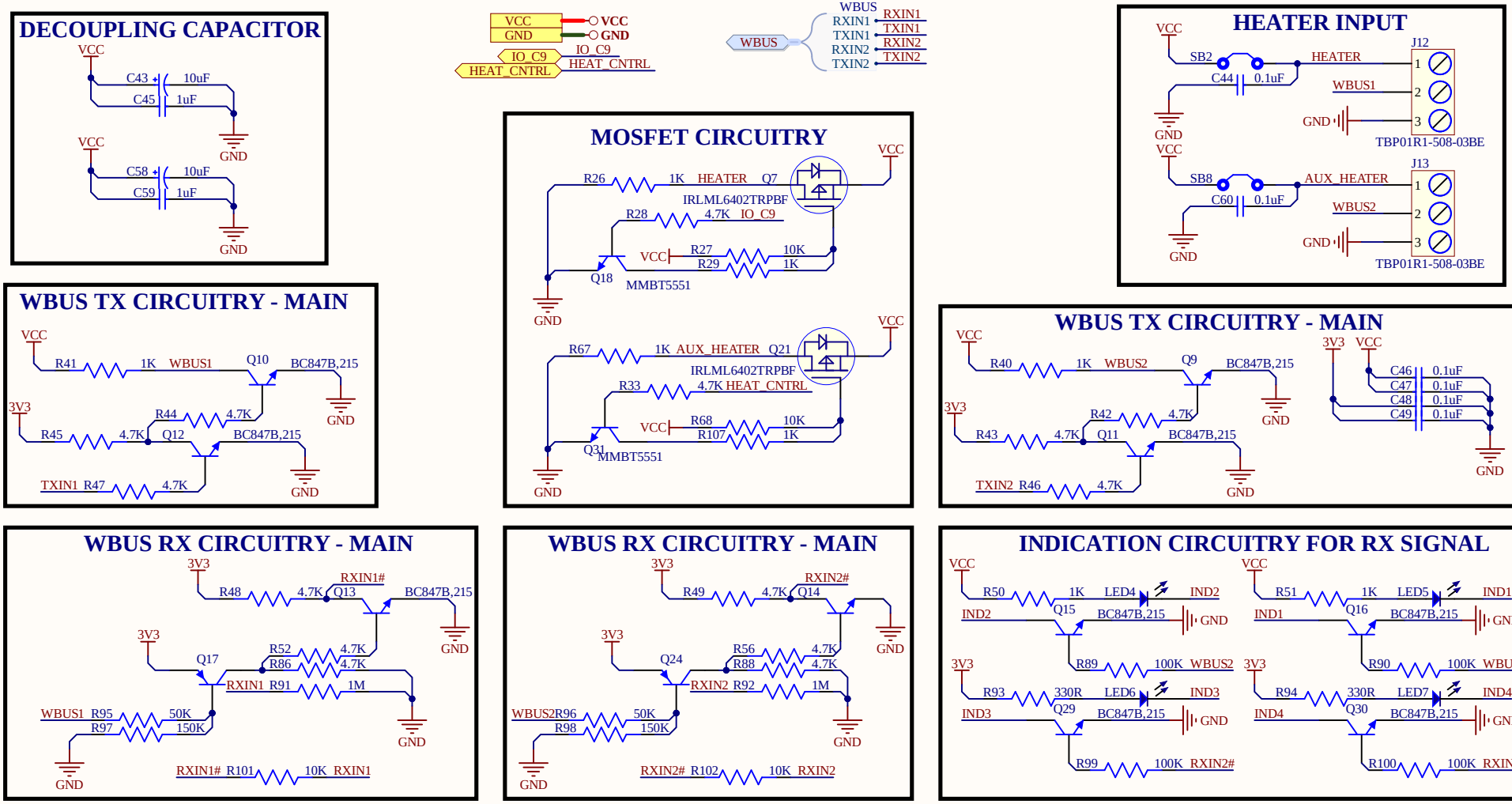

CRNT_SNS
Switches
IO_C9
GATE_DRV
VALVE_SIGNALS
SNS_SIG
LED_SIG
HEAT_CNTRL
WBUS
RELAY_SIGNALS
U_Switch_Sheet
Switch_Sheet.SchDoc
3V3
Switches
GND
U_Gate_Driver_Section
Gate_Driver_Section.SchDoc
5V
GATE_DRV
CRNT_SNS
U_Valve_Section
Valve_Section.SchDoc
VCC
VALVE_SIGNALS
RELAY_SIGNALS
U_Heater_Section
Heater_Section.SchDoc
WBUS
U_Sensor_Section
Sensor_Section.SchDoc
SNS_SIG
U_LED_Section
LED_Section.SchDoc
LED_SIG


Switches
Switches
SW_1
SW_2
SW_3
SW_4
SW_5
SW_7
SW_1
SW_2
SW_3
SW_4
SW_5
SW_7

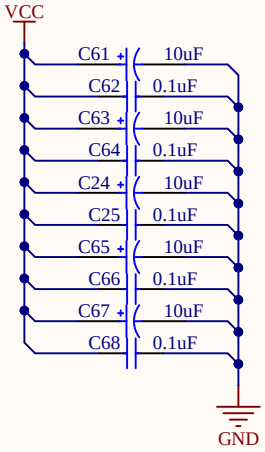

EXTERNAL INPUT - SWITCHES
3V3
R18
2.2K
SW_1
C37
0.1uF
GND
SW2
R19
SW_2
C38
SW3
R20
SW_3
C39
SW4
R21
SW_4
C40
SW5
R22
SW_5
C41
SW6
R66
SW_7
C57
SW7


3V3
GND
3V3
GND

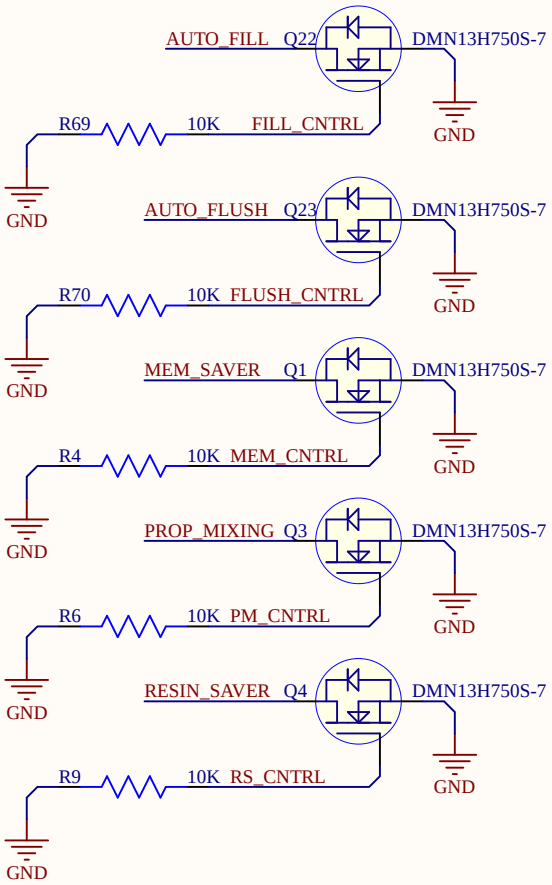

GATE_DRV
GATE_DRV
INA_1
INB_1
ENA
INA_1
INB_1
ENA
CRNT_SNS
CRNT_SNS
IP+_1
IP+_2
IP+_1
IP+_2
MOSFET DRIVERS
GND_DRV_1 Q5
IRF3205STRLPBF
DRV_OUTA1 R24
47R
GND
GND_DRV_2 Q6
IRF3205STRLPBF
DRV_OUTB1 R25
47R
GND
LEVEL TRANSLATOR
5V
U10
VCC
OUTPUT Y
INPUT A
NC
GND
INA_1
INA_1#
SN74LV1T34DBVR
GND
5V
U11
VCC
OUTPUT Y
INPUT A
NC
GND
INB_1
INB_1#
SN74LV1T34DBVR
GND
5V
U12
VCC
OUTPUT Y
INPUT A
NC
GND
ENA
ENA#
SN74LV1T34DBVR
GND
DECOUPLING CAPACITOR
5V
C55
0.1uF
GND
PULL DOWN RESISTOR
R83
200K
INA_1#
R84
200K
INB_1#
GND
INA and INB should stay pulled down if unused (Page: 2)
HEADERS FOR PUMP
J3
GND_DRV_1
IP+_1
J4
GND_DRV_2
IP+_2
LOW SIDE PUMP DRIVER IC
5V
U6
VDD
OUTA
INA
INB
OUTB
ENA
ENB
GND
INA_1#
INB_1#
DRV_OUTA1
DRV_OUTB1
5V
R85
1K
ENA#
NCP81071ADR2G
GND
3V3
5V
GND
3V3
5V
GND


DECOUPLING CAPACITOR
VCC
C43 10uF
C45 1uF
GND
VCC
C58 10uF
C59 1uF
GND
VCC
GND
IO_C9
HEAT_CNTRL
VCC
GND
IO_C9
HEAT_CNTRL
WBUS
RXIN1
TXIN1
RXIN2
TXIN2
WBUS
RXIN1
TXIN1
RXIN2
TXIN2
MOSFET CIRCUITRY
VCC
R26 1K HEATER Q7
IRLML6402TRPBF
R28 4.7K IO_C9
VCC R27 10K
R29 1K
Q18
MMBT5551
GND
VCC
R67 1K AUX_HEATER Q21
IRLML6402TRPBF
R33 4.7K HEAT_CNTRL
VCC R68 10K
R107 1K
Q31
MMBT5551
GND
HEATER INPUT
VCC
SB2
C44 0.1uF
HEATER
WBUS1
GND
J12
1
2
3
TBP01R1-508-03BE
GND
VCC
SB8
C60 0.1uF
AUX_HEATER
WBUS2
GND
J13
1
2
3
TBP01R1-508-03BE
GND
WBUS TX CIRCUITRY - MAIN
VCC
R41 1K WBUS1 Q10 BC847B,215
GND
3V3
R44 4.7K
R45 4.7K Q12 BC847B,215
GND
TXIN1 R47 4.7K
GND
WBUS TX CIRCUITRY - MAIN
VCC
R40 1K WBUS2 Q9 BC847B,215
GND
3V3
R42 4.7K
R43 4.7K Q11 BC847B,215
GND
TXIN2 R46 4.7K
GND
3V3 VCC
C46 0.1uF
C47 0.1uF
C48 0.1uF
C49 0.1uF
GND
WBUS RX CIRCUITRY - MAIN
3V3
R48 4.7K Q13 BC847B,215
RXIN1#
GND
3V3
Q17
R52 4.7K
R86 4.7K
RXIN1 R91 1M
GND
WBUS1 R95 50K
R97 150K
GND
RXIN1# R101 10K RXIN1
WBUS RX CIRCUITRY - MAIN
3V3
R49 4.7K Q14 BC847B,215
RXIN2#
GND
3V3
Q24
R56 4.7K
R88 4.7K
RXIN2 R92 1M
GND
WBUS2R96 50K
R98 150K
GND
RXIN2# R102 10K RXIN2
INDICATION CIRCUITRY FOR RX SIGNAL
VCC
R50 1K LED4 IND2
IND2
Q15 BC847B,215 GND
R89 100K WBUS2
3V3
R93 330R LED6 IND3
IND3
Q29 BC847B,215 GND
R99 100K RXIN2#
VCC
R51 1K LED5 IND1
IND1
Q16 BC847B,215 GND
R90 100K WBUS1
3V3
R94 330R LED7 IND4
IND4
Q30 BC847B,215 GND
R100 100K RXIN1#

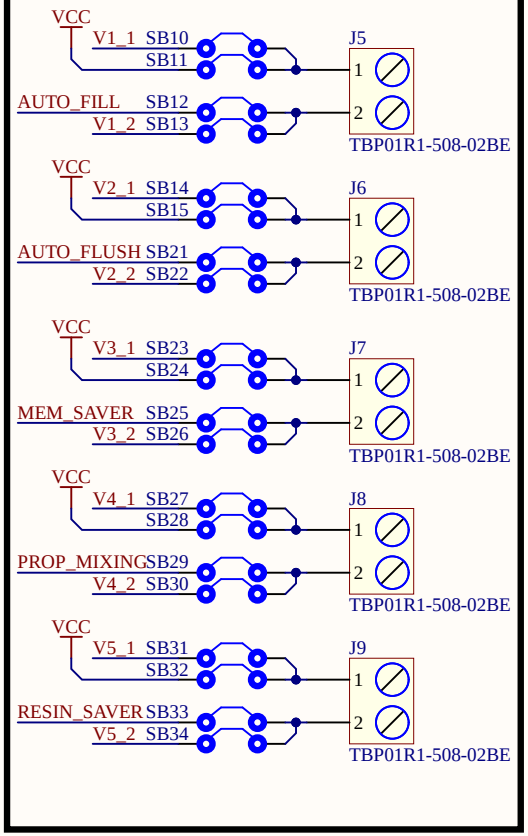
DECOUPLING CAPACITOR

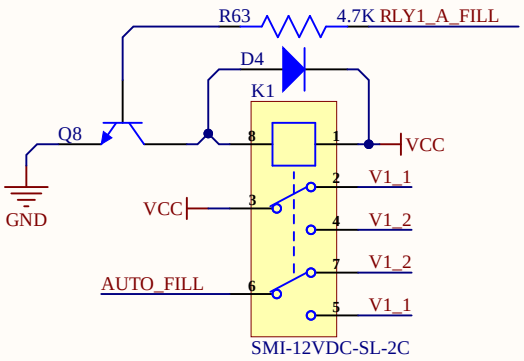


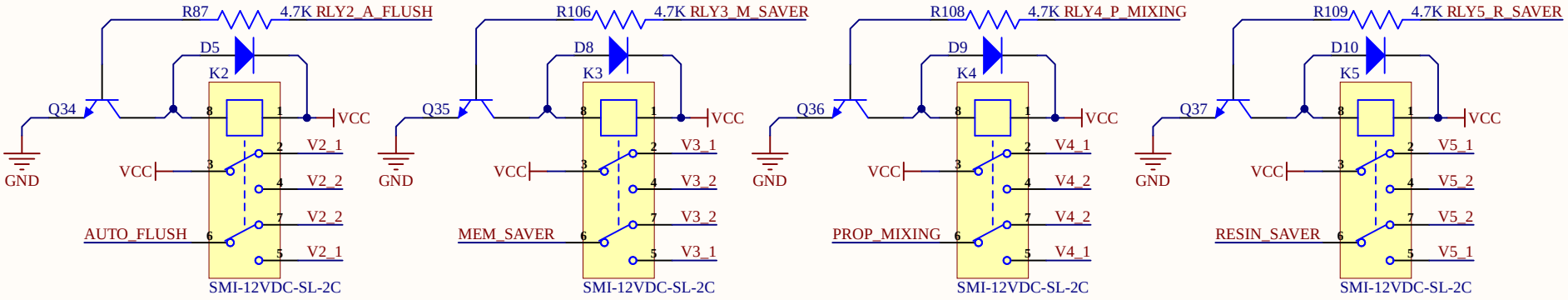
SOLENOID CONTROL CIRCUIT

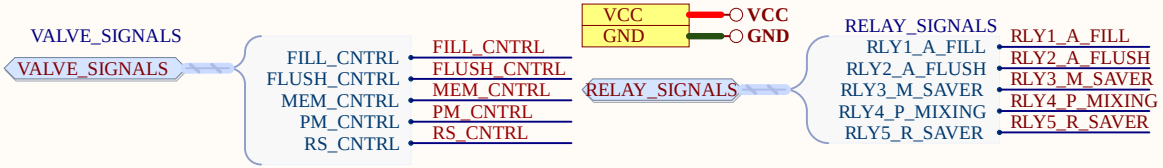


SOLENOID VALVE INPUTS

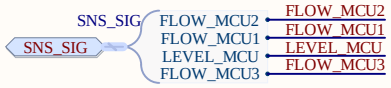




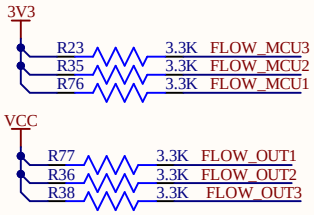




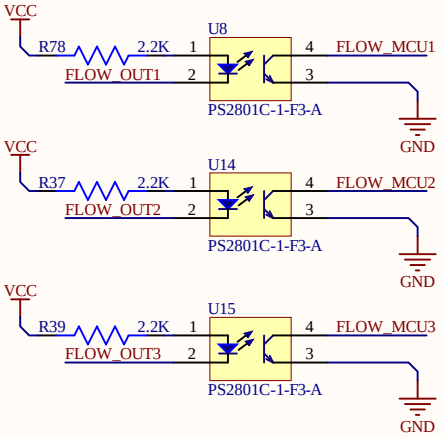
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Size A	Number	Revision
Date: 1/20/2023		Sheet of
File: C:\Users\..\Valve_Section.SchDoc		Drawn By:



PULL UP RESISTORS

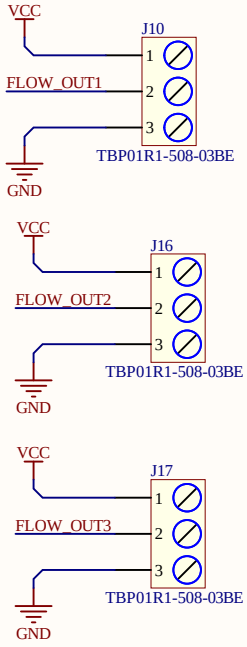


OPTO ISOLATION FOR FLOW METER



Forward voltage of diode=1.2V
With VCC=12V, IF=5.4mA
Flow meter datasheet recommends a pull up value between 3K-30K

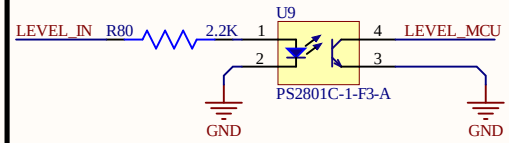
FLOW METER INPUT



SUPPORTING CIRCUIT

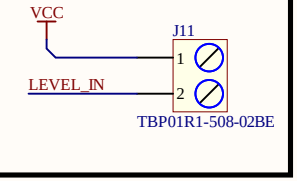


OPTO ISOLATION FOR FLOW METER



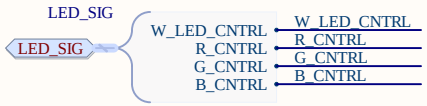
Forward voltage of diode=1.2V
With VCC=12V, IF=54mA

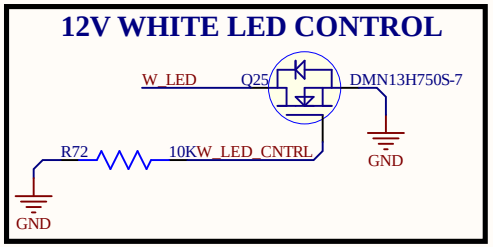
LEVEL SENSOR INPUT

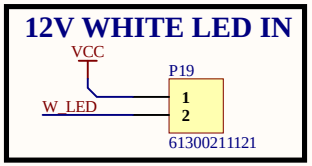


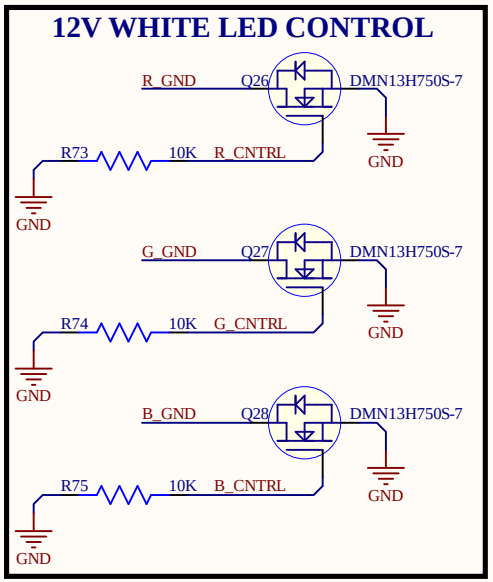


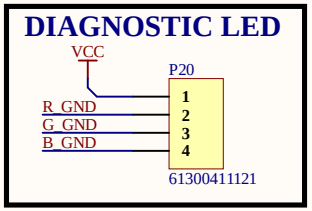
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File: C:\Users\..\Sensor_Section.SchDoc		Drawn By:





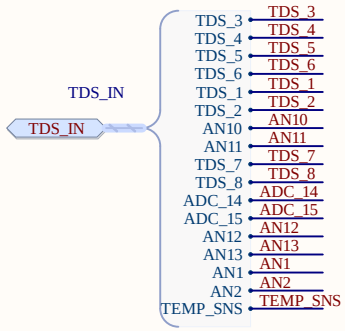




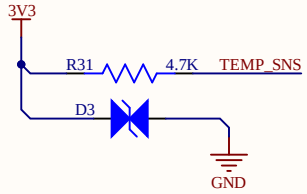




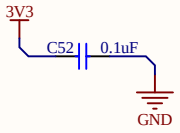
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PULL UP RESISTORS

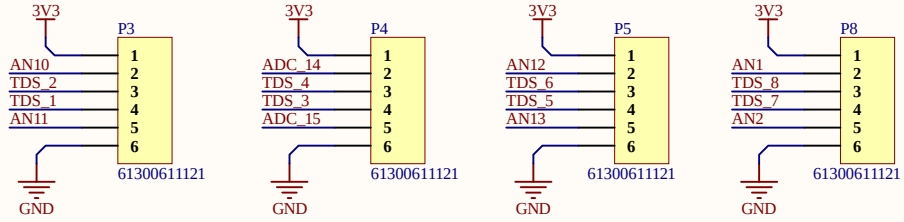


DECOUPLING CAP



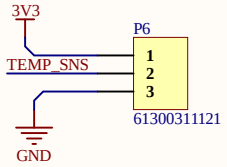


4 X TDS INPUTS

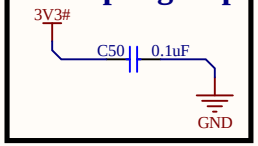


These TDS headers connect with TDS header on V3 Controller PCB via Female to Female Cable.

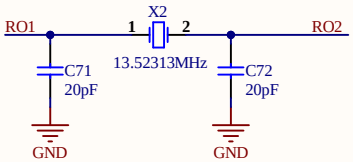
TEMP SENSOR DS18B20



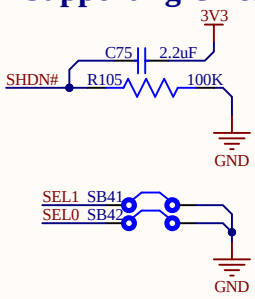
Decoupling Cap

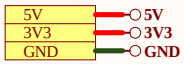


13.52127MHz Crystal



Supporting Circuit

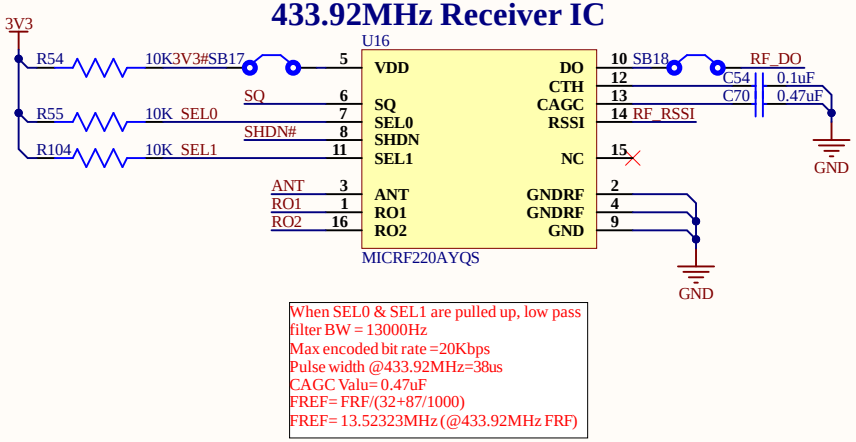






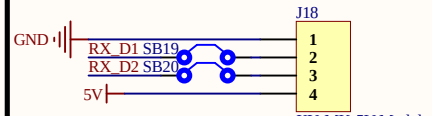


433.92MHz Receiver IC

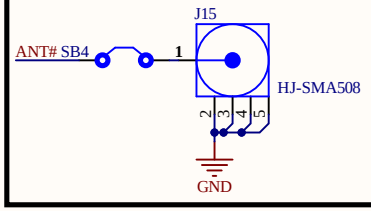


When SEL0 & SEL1 are pulled up, low pass filter BW = 13000Hz
Max encoded bit rate =20Kbps
Pulse width @433.92MHz=38us
CAGC Valu= 0.47uF
FREF= FRF/(32+87/1000)
FREF= 13.52323MHz (@433.92MHz FRF)

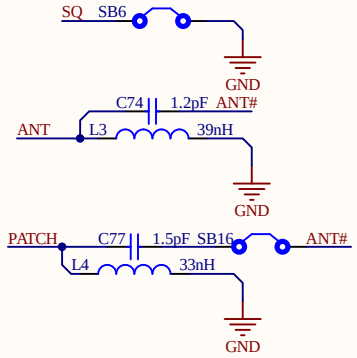
RX 433.93MHz MODULE



SMA CONNECTOR

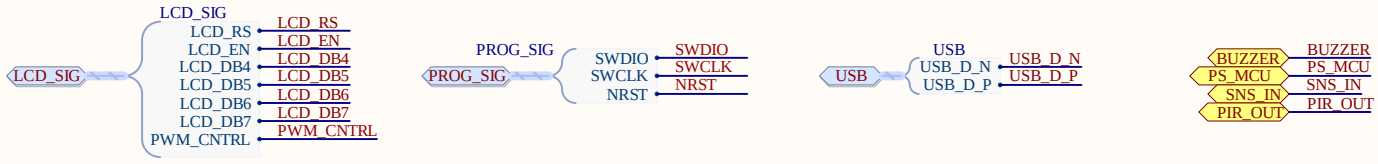


Z-MATCHING CIRCUITRY

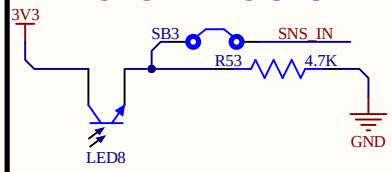


Cannot open file D:\eSpark\eSpark-Hardware\Altium Template\espark-consulting-group-150x150.jpg. File does not exist.

Title	V3_Board.PrjPcb		
Size	Number		Revision
A4	1		1
Date:	1/20/2023	Sheet 16 of	17
File:	C:\Users\..\Receiver_Section.SchDoc	Drawn By:	Huzaifa Ali

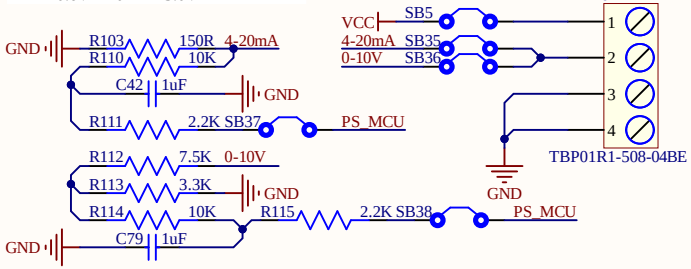


PHOTOTRANSISTOR

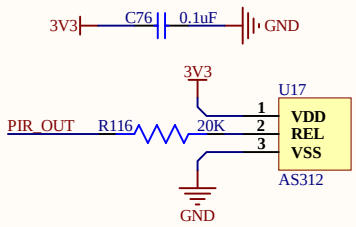


PRESSURE SENSOR

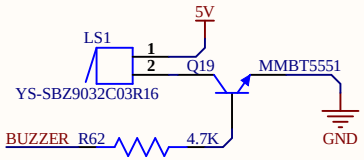
For 4-20mA respected ADC voltage will be:
4mA = 0.6V -- 20mA = 3.0V



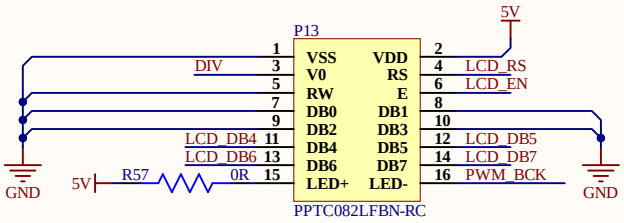
MOTION DETCTION



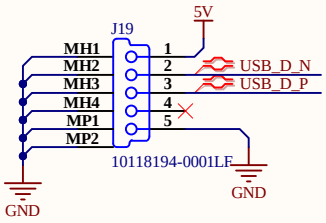
BUZZER CIRCUIT

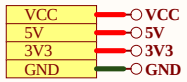


LCD CONNECTOR

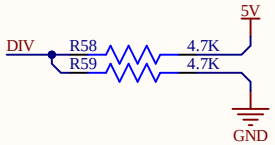


USB TYPE B CONNECTOR

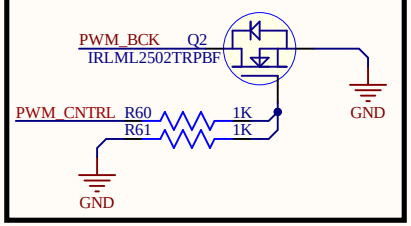




RESISTOR DIVIDER



PWM BACKLIGHT



SW DEBUGGING HEADER

